

PEMBERTON, BC, Canada

WMO: 717770

Lat: 50.306N

Lon: 122.734W

Elev: 670

StdP: 14.34

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.8	(c) 11.1	(d) -3.0	(e) 4.8	(f) 11.5	(g) 2.3	(h) 6.3	(i) 15.5	(j) 13.8	(k) 18.9	(l) 11.2	(m) 19.9	(n) 4.1	(o) 270	(p) 0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 29.1	(c) 92.4	(d) 66.1	(e) 88.4	(f) 64.6	(g) 84.3	(h) 63.2	(i) 68.9	(j) 85.6	(k) 66.5	(l) 83.3	(m) 64.6	(n) 80.4	(o) 4.9	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	89.9	73.7	60.9	81.9	70.2	59.0	76.5	67.7	33.4	84.6	31.4	83.7	29.9	80.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.6	(b) 10.3	(c) 7.8	(e) 2.0	(f) 98.3	(g) 5.1	(h) 4.2	(i) -1.6	(j) 101.3	(k) -4.6	(l) 103.7	(m) -7.5	(n) 106.1	(o) -11.2	(p) 109.1		
(a) 13.6	(b) 10.3	(c) 7.8	(e) 1.3	(f) 73.2	(g) 5.2	(h) 2.7	(i) -2.4	(j) 75.1	(k) -5.4	(l) 76.7	(m) -8.3	(n) 78.2	(o) -12.0	(p) 80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	27.9	32.2	39.8	48.0	56.1	61.4	67.4	66.5	58.5	46.3	35.0	27.9		
	DBStd	15.31	8.28	6.18	5.41	5.00	5.77	5.51	5.78	5.19	5.25	5.37	6.78	7.29		
	HDD50	2884	685	498	318	96	12	0	0	0	3	136	450	686		
	HDD65	6696	1150	918	781	511	279	138	41	47	202	578	900	1151		
	CDD50	1911	0	0	1	35	202	341	540	510	259	23	0	0		
	CDD65	250	0	0	0	0	4	29	117	92	8	0	0	0		
	CDH74	4481	0	0	0	26	275	652	1862	1387	277	2	0	0		
	CDH80	1891	0	0	0	3	71	256	887	597	77	0	0	0		
(14) Wind	WSAvg	2.5	2.2	2.2	2.4	2.9	3.1	3.2	3.1	2.6	2.0	1.9	2.1	2.2		
(15) Precipitation	PrecAvg	40.2	5.6	3.5	3.4	2.0	1.7	1.9	1.3	1.9	2.3	5.5	6.1	5.6		
	PrecMax	52.8	11.1	8.0	7.5	4.4	3.6	3.7	3.0	8.2	4.8	15.3	14.8	9.1		
	PrecMin	28.9	2.8	0.4	0.9	0.4	0.4	0.7	0.2	0.3	0.3	0.4	1.6	2.0		
	PrecStd	6.3	1.9	1.9	1.7	1.0	0.8	0.8	0.8	1.7	1.3	3.3	3.3	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	49.1	64.2	78.1	86.7	93.7	98.7	94.5	87.1	69.5	50.7	42.0		
		MCWB	38.7	41.5	48.5	55.4	60.7	66.7	68.8	66.9	63.5	56.0	49.0	40.6		
	2%	DB	38.2	44.5	56.8	69.8	81.5	87.2	93.7	91.2	81.6	62.6	47.6	38.7		
		MCWB	37.3	40.1	45.8	52.1	58.7	63.5	66.5	65.9	62.2	51.8	46.3	38.0		
	5%	DB	36.7	41.5	52.4	64.3	76.2	81.6	89.4	86.9	76.3	57.9	45.4	36.8		
		MCWB	36.1	38.3	43.8	49.3	56.8	61.5	65.2	64.6	60.2	51.4	44.4	36.3		
	10%	DB	35.5	39.0	48.8	59.6	71.0	76.1	84.8	82.5	71.2	54.4	43.3	35.2		
		MCWB	35.0	36.8	42.2	47.3	54.9	59.2	63.9	63.5	58.5	50.3	42.4	34.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.5	43.9	51.0	56.8	62.7	72.6	74.5	71.1	66.3	58.3	49.3	40.9		
		MCDB	39.7	46.8	60.6	73.6	80.4	84.9	88.7	87.2	80.9	66.8	50.1	41.3		
	2%	WB	37.7	41.0	47.4	53.4	60.0	65.4	69.4	68.3	63.7	54.8	46.6	38.3		
		MCDB	38.0	43.5	54.4	65.8	76.7	82.3	86.4	85.3	77.3	59.0	47.2	38.8		
	5%	WB	36.2	38.9	45.2	51.0	57.9	62.4	66.8	66.1	61.6	52.5	44.7	36.5		
		MCDB	36.5	40.7	50.6	61.2	72.5	78.3	84.8	82.2	73.3	56.1	45.3	36.9		
	10%	WB	35.1	37.1	43.2	48.8	56.0	60.2	64.8	64.3	59.5	50.5	42.5	35.0		
		MCDB	35.3	38.6	47.6	56.9	68.0	73.7	81.7	78.8	68.7	53.9	43.2	35.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	12.2	17.3	21.9	25.3	24.5	29.1	28.6	24.5	15.8	7.5	6.5		
		MCDBR	6.1	15.0	24.3	32.9	37.6	38.0	40.2	39.1	35.6	23.2	8.6	6.1		
		MCWBR	5.6	10.5	13.8	15.6	16.5	16.1	15.3	16.7	18.4	14.2	7.1	5.5		
		MCWBR	5.8	12.4	21.0	28.2	32.0	33.7	37.4	35.0	31.8	19.7	8.2	5.9		
(40) Clear-Sky Solar Irradiance	taud	taub	0.295	0.301	0.322	0.385	0.387	0.383	0.375	0.374	0.349	0.353	0.304	0.290		
		taud	2.203	2.230	2.188	2.022	2.181	2.321	2.371	2.399	2.462	2.364	2.327	2.196		
		Ebn at Noon	228	263	278	270	275	277	278	272	269	241	226	210		
		Edh at Noon	23	30	37	49	44	39	36	34	29	26	21	20		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	274	550	804	1294	1696	1665	1856	1624	1117	626	306	222		
		RadStd	48	83	130	164	166	203	186	121	97	102	60	38		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page